Haar Wavelets

Course: Math 414

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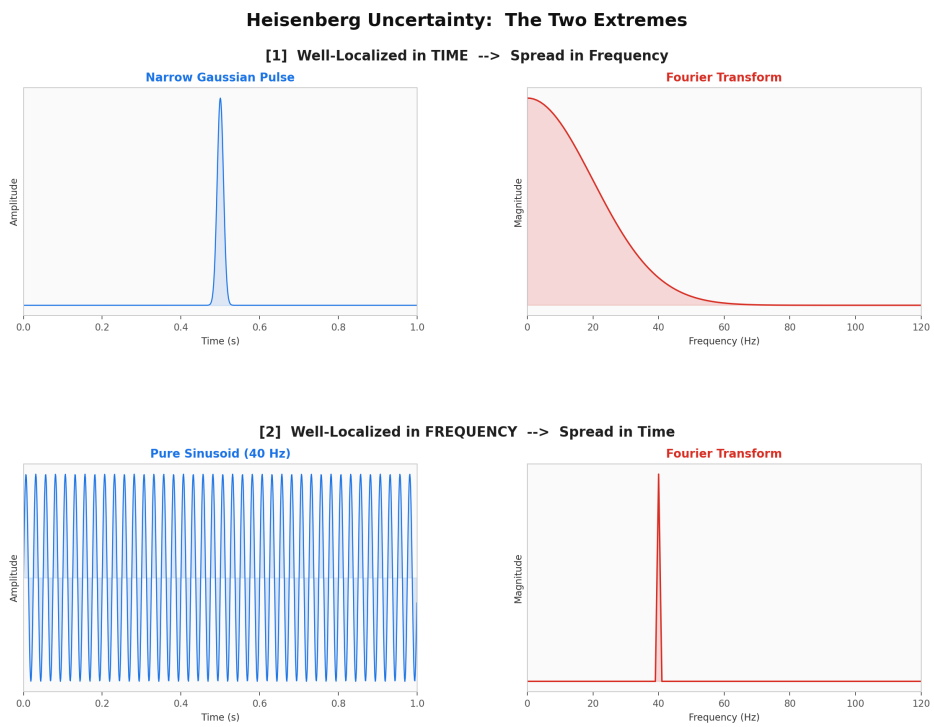
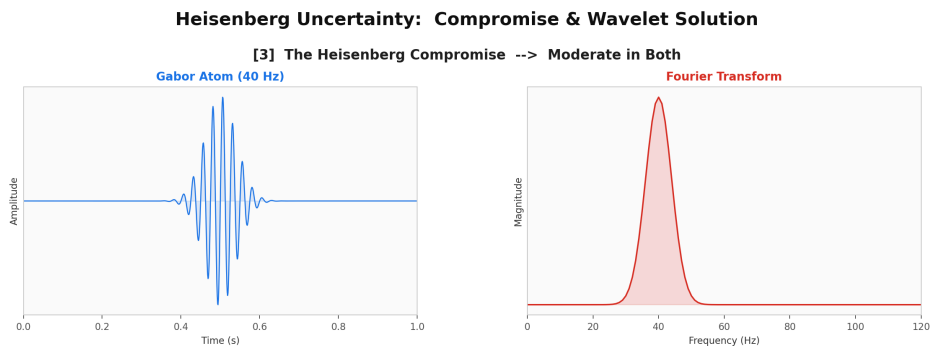
Core Idea

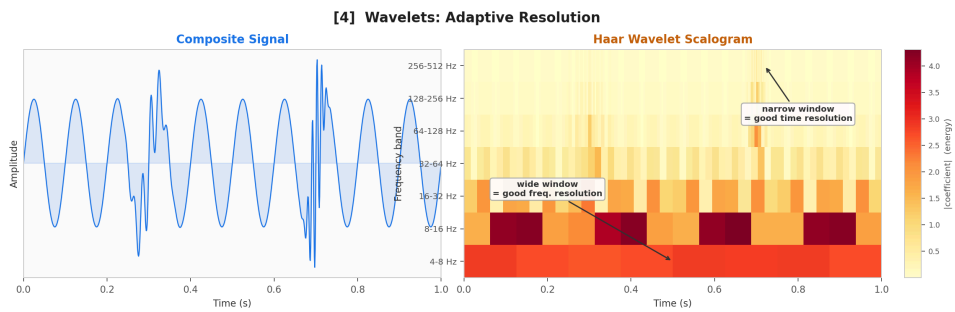
Approximate a signal using a sum of short lived waves (wavelets), instead of infinite sin and cos.

Due to Heisenberg's uncertainty principle, we cannot both be extremely accurate about location and time and location and frequency. Think about a blip of time. We know exactly when the sound happened, but know nothing about the frequency. Otherwise, if we are very certain about frequency, (such as an infinite sin wave), it has to take up the whole number line. We want to help know when an oscillation occurs.



Wavelets try to help solve this by keeping track of time and frequency information, their scale helps "zoom" in on short bursts, or "zooms out" to detect long, slow oscillations.

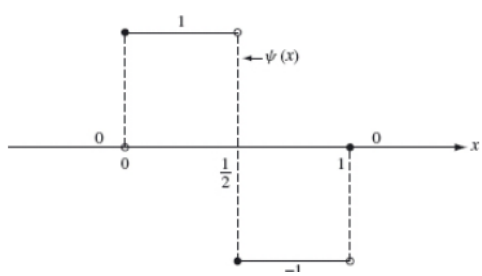




Definitions

Wavelet ψ

$$\psi(x) = \begin{cases} 1 & 0 \leq x \leq \frac{1}{2} \\ -1 & \frac{1}{2} \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$



$$W_j = \left\{ 2^{\frac{j}{2}} \psi(2^j x - k), k \in \mathbb{Z} \right\} \tag{2}$$

Properties

1. $\psi(x)$ is a member of V_1. If you think about it, it's a special case where each even half length interval has the same "shape" with one higher and then one lower line

2. $\psi(x) \perp V_0$

3. $W_0 \perp V_0$ and $V_1 = V_0 \oplus W_0$

   This means if $f \in V_1$ (already a piecewise-constant function at resolution $j = 1$), then f splits **exactly** as $f = f_0 + w_0$ where

   $$f_0 = \sum_{k \in \mathbb{Z}} a_k^0 \phi(x-k) \in V_0, \quad w_0 = \sum_{k \in \mathbb{Z}} b_k^0 \psi(x-k) \in W_0 \tag{3}$$

   Here f_0 is the coarser approximation and w_0 is the detail lost when going from resolution 1 to 0. This is exact because f was already in V_1. The scaling relations notes derive the explicit coefficients: $a_k^0 = \frac{1}{2}(a_{2k}^1 + a_{2k+1}^1)$ and $b_k^0 = \frac{1}{2}(a_{2k}^1 - a_{2k+1}^1)$.

4. More generally $V_{j+1} = V_j \oplus W_j$

5. For an arbitrary $f \in L^2(\mathbb{R})$ (not necessarily in any V_j), we need infinitely many detail layers. f_0 is the projection onto V_0, and each w_j captures the detail at scale j:

   $$f = f_0 + \sum_{j=0}^{\infty} w_j \tag{4}$$

Direct sum. This means every element of V_1 can be made up of unique combo of v_0 and w_o

f_0 is defined in scaling as the projection at resolution $j = 0$

Proofs

Proving $V_{j+1} = V_j \oplus W_j$

We must show two facts:

1. Every function in W_j is orthogonal to every other function in V_j.
2. Any function in V_{j+1} orthogonal to V_j must belong to W_j. Since $W_j \wedge V_j$ are orthogonal (if we prove the above), proving this means we can write anything as a sum of v_j, w_j.